

FACULTY OF ENGINEERING, UNIVERSITY OF JAFFNA
EC4010-DIGITAL DESIGN
ASSIGNMENT-02
ALU

Assignment: Understanding the ALU and Its Operations

Objective: This assignment aims to help you understand the role of the Arithmetic Logic Unit (ALU) in digital systems. You will explore the various operations an ALU can perform, their significance, and how they are implemented.

Question 1: Define the Arithmetic Logic Unit (ALU)

Question 2: List and Explain the Key Operations Performed by an ALU

Question 3: Design a Simple ALU

Question 4: Use Cases for ALU Operations

Submission Instructions:

- Answer all the questions thoroughly.
- Provide detailed explanations and examples where necessary.
- Provide references to any materials or external sources used.
- Submission Date:07/02/2025